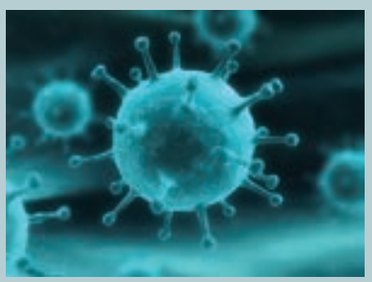
Nipah virus is a zoonotic virus (transmitted from animals to humans) and also transmitted through contaminated food. In infected people, it causes a range of illness from asymptomatic infection to acute respiratory illness and fatal encephalitis.



living cells. Outside the living cells they cannot carry out the characteristics of a living organism. Viruses invade living cells, forcing the cells to create new viruses. The new viruses break out of the cell, killing it and invade other cells in the body, causing diseases in human beings. *Rhino viruses* cause one of the most infectious human ailment called the "**Common cold**".

Viral diseases are generally grouped into four types on the basis of the symptoms produced in the body organs.

i. Pneumotropic diseases (respiratory tract infected by influenza)

ii. Dermotropic diseases (skin and subcutaneous tissues affected by chicken pox and measles)

iii. Viscerotropic diseases (blood and visceral organs affected by yellow fever and dengue fever)

iv. Neurotropic diseases (central nervous system affected by rabies and polio). Some common viral diseases of human beings are given in **table 7.2**.

7.1.2 Protozoan diseases

About 15 genera of protozoans live as parasites within the human body and cause diseases.



Swine flu was first recognised in the 1919 pandemic and still circulates as a seasonal flu virus. Swine flu is caused by the H1N1 virus strain. Symptoms include fever, cough, sore throat, chills, weakness and body aches. Children, pregnant women and the elderly are at risk from severe infection.

Amoebiasis also called amoebic dysentery or amoebic colitis is caused by *Entamoeba histolytica,* which lives in the human large intestine and feeds on mucus and bacteria **(Fig. 7.1)**. Infective stage of this parasite is the **trophozoite**, which penetrates the walls of the host intestine (colon) and secretes histolytic enzymes causing ulceration, bleeding, abdominal pain and stools with excess mucus. Symptoms of amoebiasis can range from diarrhoea to dysentery with blood and mucus in the stool. **House flies** (*Musca domestica*) acts as a carrier for transmitting the parasite from contaminated faeces and water.

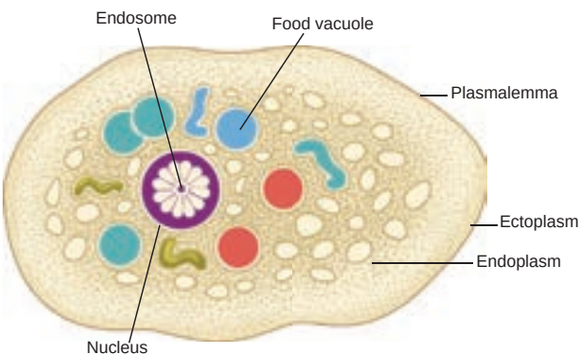


Fig. 7.1 *Entamoeba histolytica*

African sleeping sickness is caused by *Trypanosoma species*. *Trypanosoma* is generally transmitted by the blood sucking **Tsetse** flies. Three species of *Trypanosoma* cause sleeping sickness in man.

1. *T. gambiense* is transmitted by *Glossina palpalis* **(Tsetse fly)** and causes Gambian or Central African sleeping sickness **(Fig. 7.2).**

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